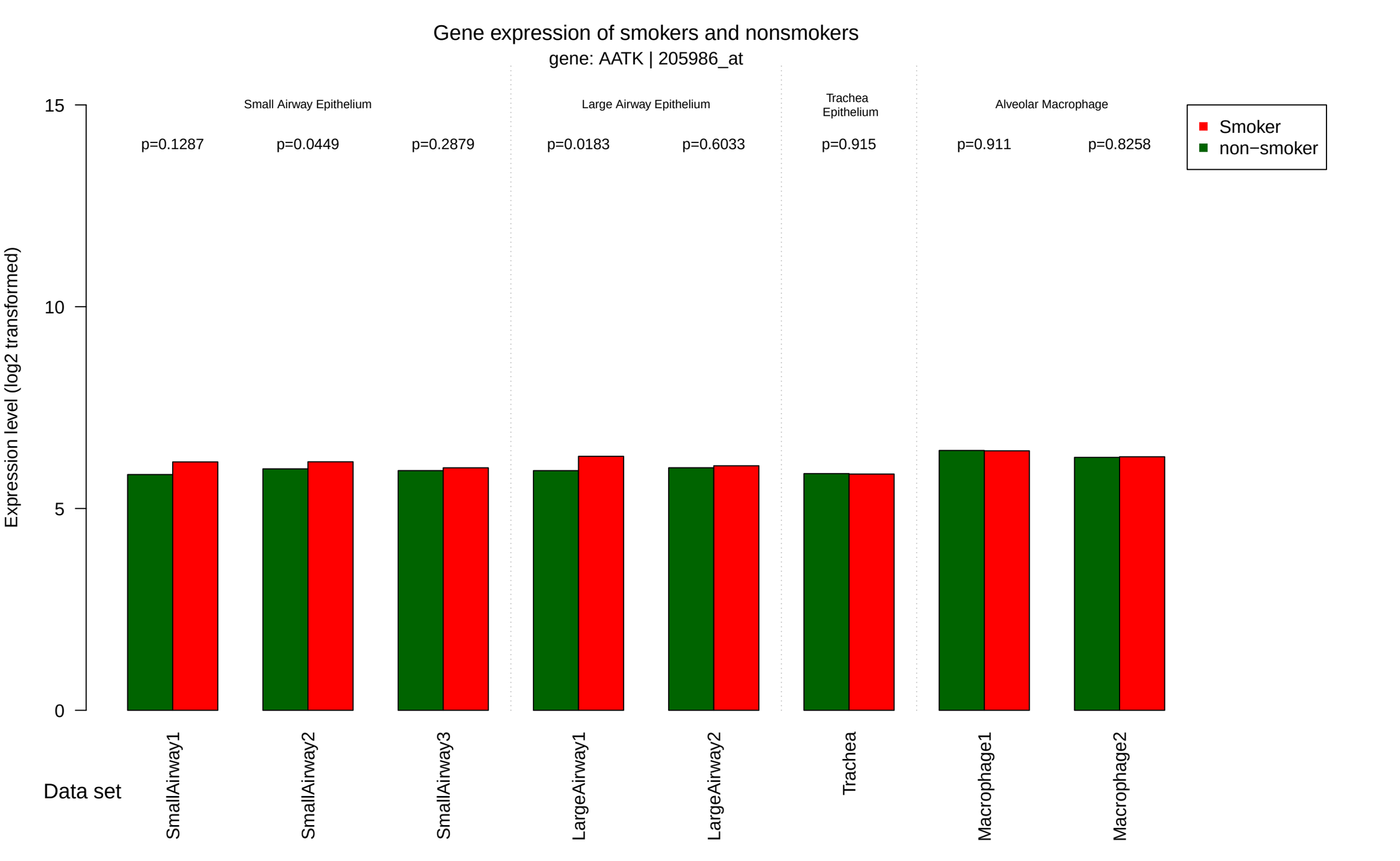
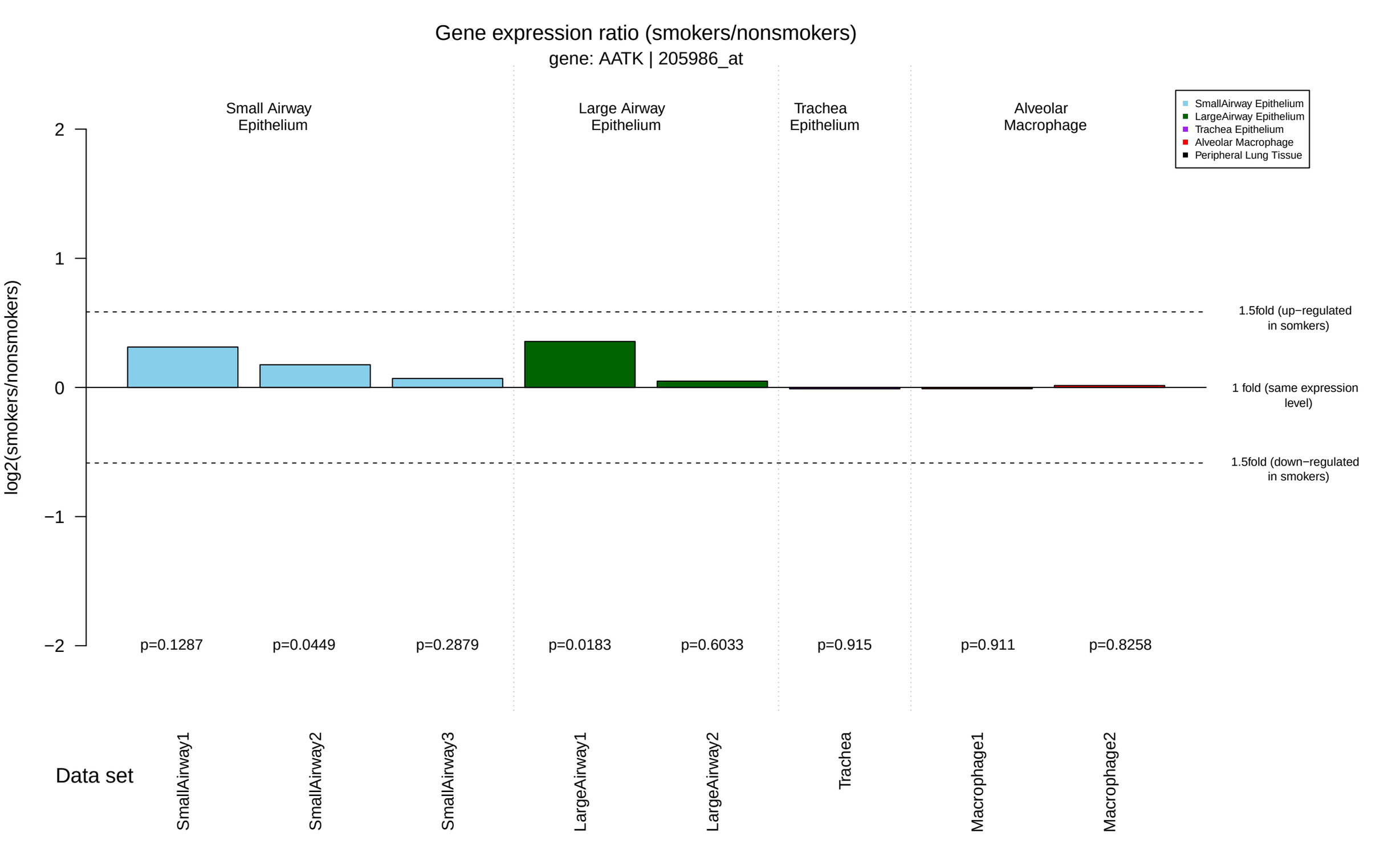
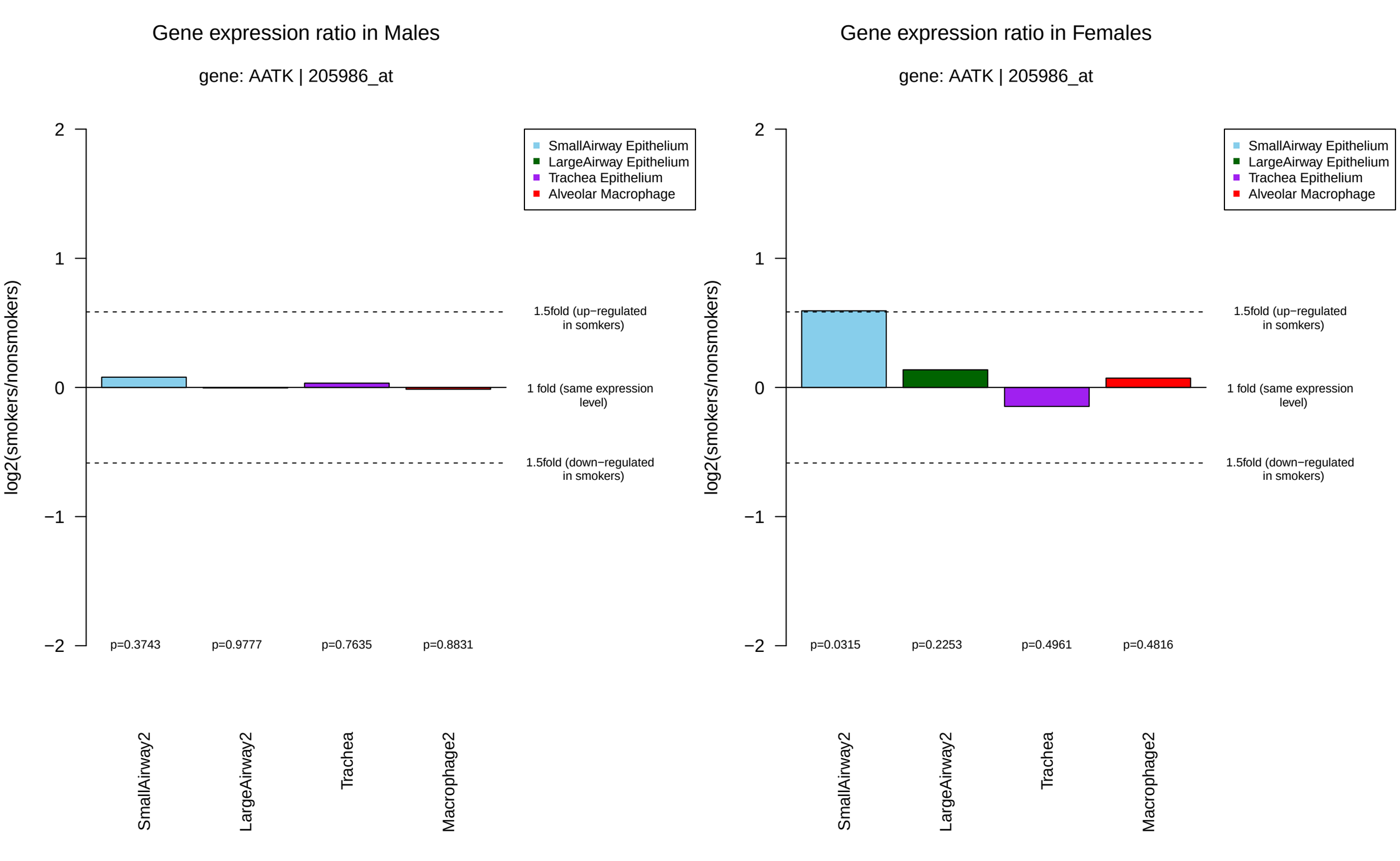
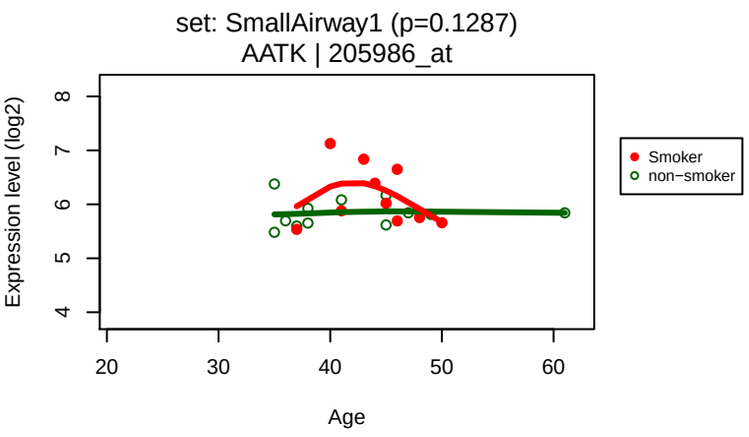
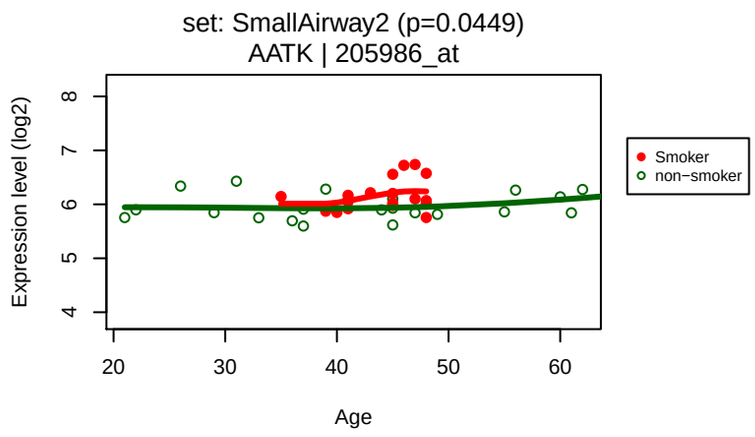
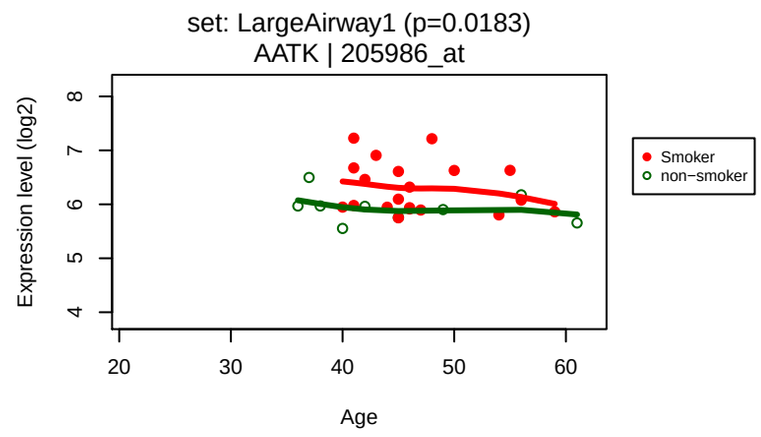

Gene expression of smokers and nonsmokers
gene: AATK | 205986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1287
p=0.0449
p=0.2879
p=0.0183
p=0.6033
p=0.915
p=0.911
p=0.8258
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

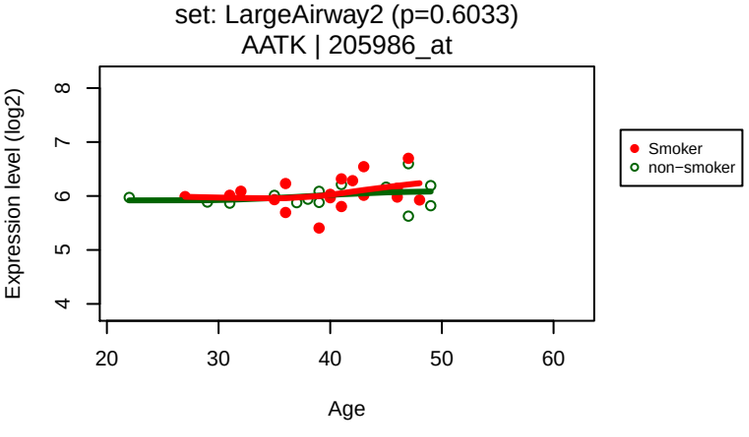

Gene expression ratio (smokers/nonsmokers)
gene: AATK | 205986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1287
p=0.0449
p=0.2879
p=0.0183
p=0.6033
p=0.915
p=0.911
p=0.8258
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

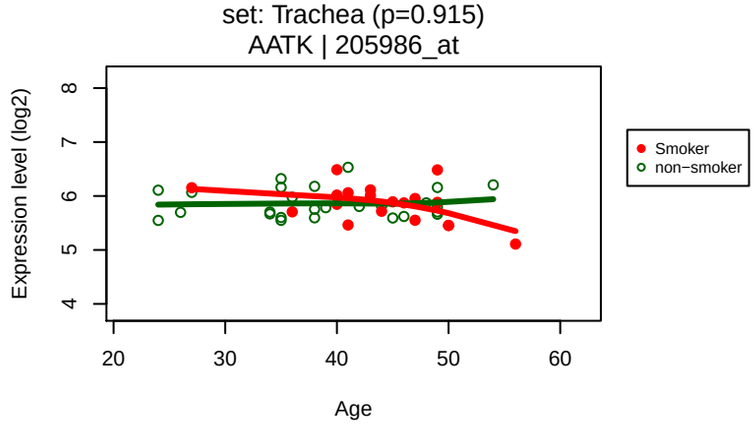

Gene expression ratio in Males
gene: AATK | 205986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3743
p=0.9777
p=0.7635
p=0.8831
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AATK | 205986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0315
p=0.2253
p=0.4961
p=0.4816
SmallAirway2
LargeAirway2
Trachea
Macrophage2

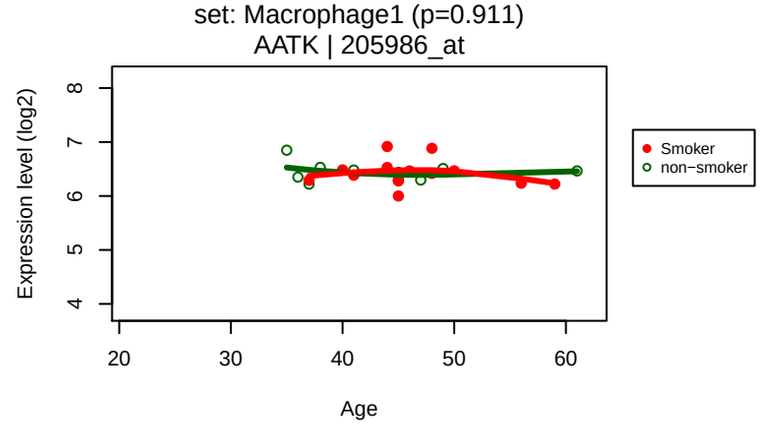

set: SmallAirway1 (p=0.1287)
AATK | 205986_at
Expression level (log2)
Age
Smoker
non−smoker

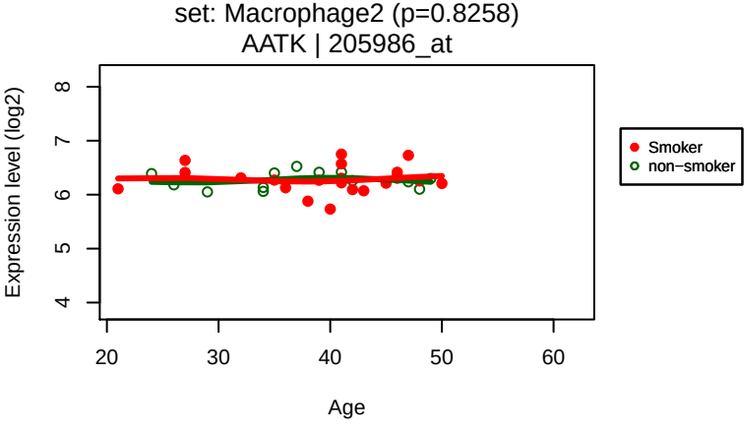

set: SmallAirway2 (p=0.0449)
AATK | 205986_at
Expression level (log2)
Age
Smoker
non−smoker

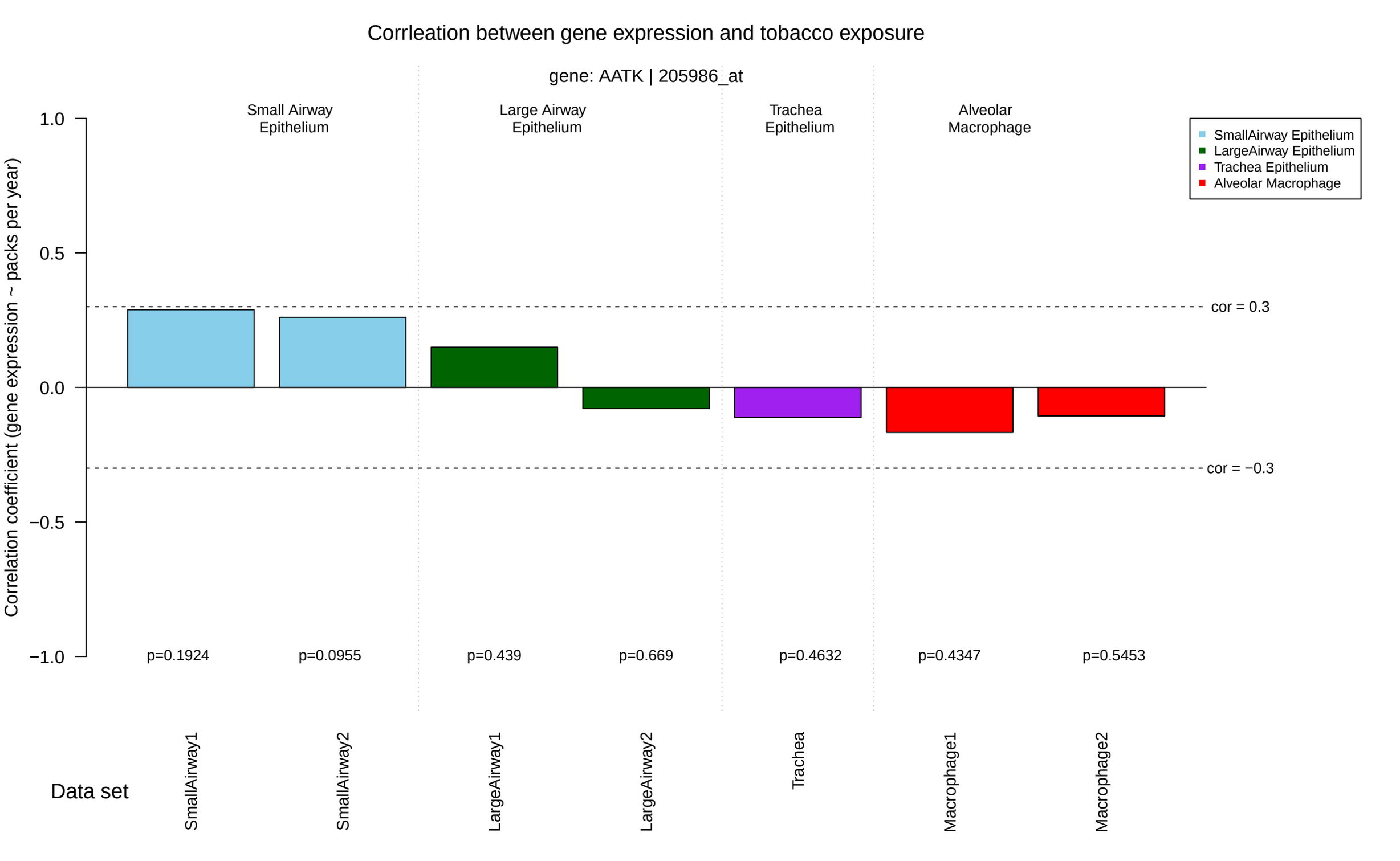

set: LargeAirway1 (p=0.0183)
AATK | 205986_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6033)
AATK | 205986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.915)
AATK | 205986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.911)
AATK | 205986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8258)
AATK | 205986_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AATK | 205986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1924
p=0.0955
p=0.439
p=0.669
p=0.4632
p=0.4347
p=0.5453
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2